Probability and Estimation Theory for Aerospace Engineering, 3 credits

1 Course information:

Estimation theory plays a fundamental role in robotics, data science, Google Pagerank, GPS system, ect. This course presents an introduction to probability and estimation theory at the graduate/undergraduate level to develop a deep understanding of estimation theory and some of its applications. The course presents many of the classical distributions, Binomial, Gaussian, Poisson, exponential, ect. Then the students study estimation theory with applications to filtering, Pagerank and some classical conditional expectation problems. A special emphasis is places on multivariable Gaussian densities and estimation.

2 Course Description

- Probability measure, conditional probability and Bayes rule
- The gamblers ruin problem
- Pagerank and Google's search algorithm
- Random variables, distribution and density functions
- The exponential, uniform and Gaussian random variables
- The binomial, Poisson random variables.
- Functions of a random variable
- Joint densities and distributions
- The Raleigh density
- The Box-Muller transformation
- Expectation, mean, variance standard deviation, and median
- Applications to Gaussian, exponential, binomial etc.
- Dream and cheating in Minecraft.
- The projection theorem and applications to least squares estimation
- Condition expectation
- Condition expectation of Gaussian random vectors and the orthogonal projection
- The central limit theorem
- Maximal likelihood estimation

3 Learning Resources and Reference Text

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook: The textbook is a set of notes. The notes are posted on Brightspace in a .pdf file free of charge.

4 Assignments

The course assigns homework which is graded and included in part of the final grade. There may also be one or two special projects in the course. The homework will be handed in by gradescope or brightspace. The solutions for the homework will be provided.

Late homework will be accepted up to 48 hours after the due date and time, but a 10% penalty will incur for every 12 hours it is late. In other words:

- 0-12 hours late = 10% deduction
- 12-24 hours late = 20% deduction
- 24-36 hours late = 30% deduction
- 36-48 hours late = 40% deduction.

After 48 hours, no late homework will be accepted unless there is a medical or university related reason; the late due date may be adjusted in such a case.

5 Course Learning Objectives

On completing this course the student shall be able to:

1. Use probability theory to solve elementary problems involving Bayes rule.
2. To use Pagerank to analyze and rank data.
3. Recognize when two events are independent.
4. To know how to use and apply all the elementary distributions including binomial, Poisson, uniform, exponential, Gaussian and Cauchy.
5. To have a good understanding of the median, mean and standard deviation and their corresponding optimization problems.
6. Use probability theory to analyze data.

7. To have a relatively "deep" understanding of conditional expectation and its application to estimation theory.

8. To have the background and prerequisites for a course on Kalman filtering.

6 Grading scale

- Grades are A, B, C, D, F and no $\pm$ grades.
- The homework counts for 35% of the final grade. The exams and final exam will account for 65% of the final grade.
- The final grade is determined by a curve. The instructor sorts all the grades and then calculates the mean, median and standard deviation. Based on this data an A is a minimum of 35% of the class (and could be a higher percentage). The rest of the grades are B with the exception of usually a very few students which did very little for the course. They may receive a C, D or F grade.
- The grading and distribution of grades in this course are done totally blind. The instructor has no idea what student is associated with any score when assigning grades.